Engineering résumé

LinkedIn | GitHub

Education

Bachelor of Science in Electrical Engineering — in progress

Relevant coursework: Digital System Design, Object-Oriented Programming, Discrete Math, Linear Algebra

Experience

Undergraduate Student Researcher — 3D FPGA research

Aug 2026 – Present

- Prototyped a 3D configurable logic block with 10 LUTs split 5/5 across two die layers; extended VTR architecture parsing and placement and OpenFPGA RTL generation.
- Took a single-die CLB baseline through Nangate45 synthesis and OpenROAD place-and-route; verified RTL, synthesized, and post-route netlists with a 44-case functional simulation suite.
- Traced an RTL generation failure affecting 3 of 6 LaZagna architecture types to an 88-bit versus 86-bit interlayer port mismatch, and fixed a clean-clone build failure.

ASIC Physical Design Team Member — student design team

Sep 2026 – Present

- Designed an RV32I-subset CPU in SystemVerilog and verified it using a class-based testbench, reference model, scoreboard, and SystemVerilog Assertions in Cadence Xcelium.
- Reached 98.9% DUT code coverage in Cadence IMC with a 34-case directed test plan and constrained-random stimulus.
- Took the CPU and two SRAM macros from RTL to routed GDS in Sky130 with Synopsys Design Compiler and Cadence Innovus; sized a 1253 × 626 µm core for 50% utilization and placed macros with halos and a block power ring.
- Wrote Tcl timing-analysis automation over 1,100+ paths and traced the worst setup path to approximately 33 cells through decode, the register file, and ALU.
- Added 33 integrated clock gates and high-speed cell swapping; ran DRC/LVS in Cadence Pegasus and traced all 35 LVS shorts to merged tie-off nets on clock-gate test-enable pins.

IT Intern — medical technology

May 2025 – Aug 2025

- Migrated approximately 20 workstations to Windows 11 and configured hardware firewalls to enforce network security policy.

Projects & technical skills

GPU-Accelerated Logic Gate Simulator

CUDA · C++ · Yosys

- Built a gate-level simulator for a 20,170-gate, 8,676-flop CPU netlist, using Kahn’s algorithm for levelization and integer arrays for GPU memory access.
- Cut GPU runtime by 4.7× with one persistent CUDA kernel using Cooperative Groups and grid-stride loops; reached approximately 2× CPU speed at 100,000 cycles on an RTX 3060 Ti.
- Fixed nondeterministic results caused by flattened wire-name collisions with a Yosys autoname pass; verified a bit-exact CPU/GPU match on every signal’s final state.

16-Bit Pipelined RISC CPU

SystemVerilog · Python · Gowin FPGA

- Designed a five-stage processor with a custom 12-instruction ISA, Python assembler, forwarding, load-use stalls, and branch flushing.
- Deployed on a Tang Nano 9K, showing Fibonacci numbers on two TM1637 displays using a double-dabble binary-to-BCD converter and a custom serial driver.
- Debugged a synthesis crash caused by a 65,536-way multiplexer and memory left uninitialized by a simulation-only initial block.

Posture Tracking Wearable

C++ · ESP32 · BLE · Rails · TimescaleDB

- Owned hardware and firmware on a four-person team building an ESP32 wearable with on-device slouch detection and a web dashboard.
- Brought up a BNO055 IMU over I²C and implemented a calibrated upright baseline, ±180° wraparound handling, and a persistence window to ignore brief movement.

Technical skills

Languages & APIs: SystemVerilog, Verilog, C, C++, CUDA, Python, Tcl, MATLAB, Java

EDA: Cadence Innovus, Xcelium, SimVision, IMC, Pegasus, Virtuoso; Synopsys Design Compiler; OpenROAD, Yosys, VTR, OpenFPGA, GTKWave

Concepts: RTL design, VLSI, physical design, static timing analysis, clock gating, DRC/LVS, computer architecture, RISC-V ISA, FSMs, constrained-random verification, SystemVerilog Assertions

Hardware & tools: Gowin Tang Nano 9K, Gowin EDA, ESP32, SPI, I²C, Git, Linux